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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

StudentFeePayment Tracking and Receipt Generation System

Submitted by

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Semester: **II**

DECLARATION

I hereby declare that the project entitled "**Student Fee Payment Tracking and Receipt Generation System"** submitted in partial fulfillment of the requirements for the course **Java Programming** is a **record of my original work** carried out under the guidance of **Mr.Dharaneesh,** Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Anjali M

CERTIFICATE

This is to certify that the project entitled **"Student Fee Payment Tracking and Receipt Generation System"** is a **bona fide work** carried out by **Anjali M [1-C]**, **[B.SC. Computer Science]**, **KG College of Arts & Science.**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:10/04/2026

Submitted for the Viva-Voce Examination held on 10/04/2026

Guide:
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ABSRACT

The Student Fee Payment Tracking and Receipt Generation System is a software application designed to efficiently manage and monitor student fee transactions in educational institutions. The system aims to replace traditional manual record-keeping methods with a computerized solution that ensures accuracy, transparency, and easy access to financial data.

This system allows administrators to record student details, track fee payments, and maintain up-to-date records of pending and completed transactions. It also provides the functionality to automatically generate receipts for each payment, reducing errors and saving time. The system ensures secure storage of data and allows quick retrieval of information when needed.

Additionally, the application helps in minimizing human errors, improving efficiency, and enhancing overall management of student fee records. It is user-friendly and can be used by staff with basic computer knowledge. The system plays a vital role in streamlining financial operations and maintaining proper documentation within educational institution.

1. INTRODUCTION

In educational institutions, managing student fee payments manually is time-consuming and error-prone. Staff often use paper records or spreadsheets, which may lead to calculation mistakes and difficulty in tracking payments.The Student Fee Payment Tracking System is developed to solve these issues by automating fee management. It allows easy entry of student details, fee calculation, and receipt generation.This system improves accuracy, reduces workload, and ensures proper record maintenance.increasingly adopting computerized systems to manage their day-to-day operations efficiently. One of the most important administrative tasks in schools and colleges is managing student fee payments. Traditionally, many institutions rely on manual methods such as paper records or basic spreadsheets to track fee transactions. However, these methods are often time-consuming, error-prone, and difficult to maintain, especially when dealing with a large number of students.

1.1 Objective

To develop a computerized system for managing student fee payment records efficiently.

1. To maintain accurate and up-to-date details of student information and fee transactions.
2. To track paid and pending fees of students easily.
3. To generate fee payment receipts automatically.
4. To reduce manual work and minimize human errors in fee management.
5. To ensure secure storage and quick retrieval of data.
6. To improve overall efficiency and transparency in the fee collection process.

1.2 Overview

The *Student Fee Payment Tracking and Receipt Generation System* is a software application developed to simplify the process of managing student fee payments in educational institutions. Traditionally, fee records are maintained manually, which can lead to errors, data loss, and time-consuming processes. This system provides a digital solution to overcome these challenges.

The application allows administrators to store student details, record fee payments, and monitor outstanding dues efficiently. It also enables automatic generation of payment receipts, which helps in maintaining proper documentation and reducing manual effort. The system ensures that all financial data is organized and easily accessible whenever required.

1.3 Features

Student Management
Stores and manages complete student details such as name, ID, course, and contact information.

- **Fee Payment Tracking**
  Keeps track of all fee transactions including paid and pending amounts.
- **Automated Receipt Generation**
  Generates receipts instantly after each payment with details like date, amount, and student information.
- **Search and Retrieve Records**
  Allows quick searching of student records using ID or name.
- **Update and Delete Records**
  Enables modification or removal of student and payment details when required.
- **User-Friendly Interface**
  Simple and easy-to-use system for staff with basic computer knowledge.
- **Data Accuracy and Security**
  Reduces human errors and ensures safe storage of data.
- **Report Generation**
  Generates reports for fee collection, pending dues, and transaction history.
- **Time-Saving Process**
  Automates manual work, reducing time and effort.

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied carefully and system requirements are identified. It helps in understanding what the system should do and how it can be designed to solve real-world problems efficiently.In the case of the Student Fee Payment Tracking

and Receipt Generation System, program analysis focuses on identifying the issues

2.1 Problem Definition

In many educational institutions, student fee management is handled manually using paper records or simple spreadsheets. This traditional approach creates several problems:

*Errors in fee calculation due to manual work

*Time-consuming process of recording and verifying payments

*Difficulty in tracking paid and pending fees

*Chances of losing important records

*Delay in generating fee receipts

*Because of these issues, there is a strong need for a computerized system that can automate fee tracking and receipt generation.

2.2 Project Overview

The Student Fee Payment Tracking and Receipt Generation System is a Java-based application designed to simplify fee management in educational institutions.

The system provides the following functionalities:

Stores student details such as name, ID, and course

*Accepts fee payment details from the user

*Calculates total fee, paid amount, and remaining balance

*Tracks payment status (paid / pending)

*Generates a structured receipt automatically

*Displays all details clearly for verification

2.3 Scopes and Limitations

Scope

The scope defines where and how the system can be used:

*Can be used in schools, colleges, and training institutes

*Helps in managing student fee records digitally

*Useful for administrative departments

*Can generate receipts quickly and accurately

*Can be expanded with database integration

limitations

*The limitations of the current system include:

*Does not support online payment methods

*Limited to basic functionality

*Cannot handle multiple users at the same time

*No advanced security features

*No automatic report generation

3. SYSTEM DESIGN

The *Student Fee Payment Tracking and Receipt Generation System* is designed using a structured approach to ensure smooth functioning and easy maintenance. The system consists of the following components:

1.Input design

- Collects data from the user (admin/staff).
- Inputs include:
    - Student details (ID, name, course, contact)
    - Fee details (amount, date, payment mode)
- Ensures validation to avoid incorrect or incomplete data entry.

2.Output design

- Displays processed information in a clear format.
- Outputs include:
    - Payment receipts
    - Student fee status (paid/pending)
    - Reports of transactions
- Receipts can be printed or saved.

3. Database Design

- Stores all the system data securely.
- Main tables:
    - Student Table (Student ID, Name, Course, Phone)
    - Fee Table (Student ID, Amount Paid, Balance, Date)
    - Receipt Table (Receipt No, Student ID, Payment Details)
- Ensures quick access and data consistency.

4. Process Design

- Handles operations like:
    - Adding new student records
    - Recording fee payments
    - Calculating balance fees
    - Generating receipts automatically
- Ensures smooth data flow between input and output.

5. User Interface Design

- Simple and interactive interface.
- Easy navigation for adding, updating, and viewing records.
- Designed for non-technical users.

6. System Architecture

- Follows a basic three-layer architecture:

    - Presentation Layer (User Interface)
    - Application Layer (Processing Logic)
    - Database Layer (Data Storage)
- Ensures modularity and easy maintenance.

4. METHODOLOGY

Methodology describes the step-by-step process followed to develop the project. It explains how the system is planned, designed, implemented, tested, and finalized. A proper methodology ensures that the project is developed in an organized and systematic manner.

For the Student Fee Payment Tracking and Receipt Generation System, a structured approach is followed to ensure that the system meets user requirements and functions correctly.

4.1. Problem Analysis

In many educational institutions, student fee management is still handled manually using registers or basic spreadsheets. This traditional approach leads to several challenges and inefficiencies in maintaining accurate financial records.

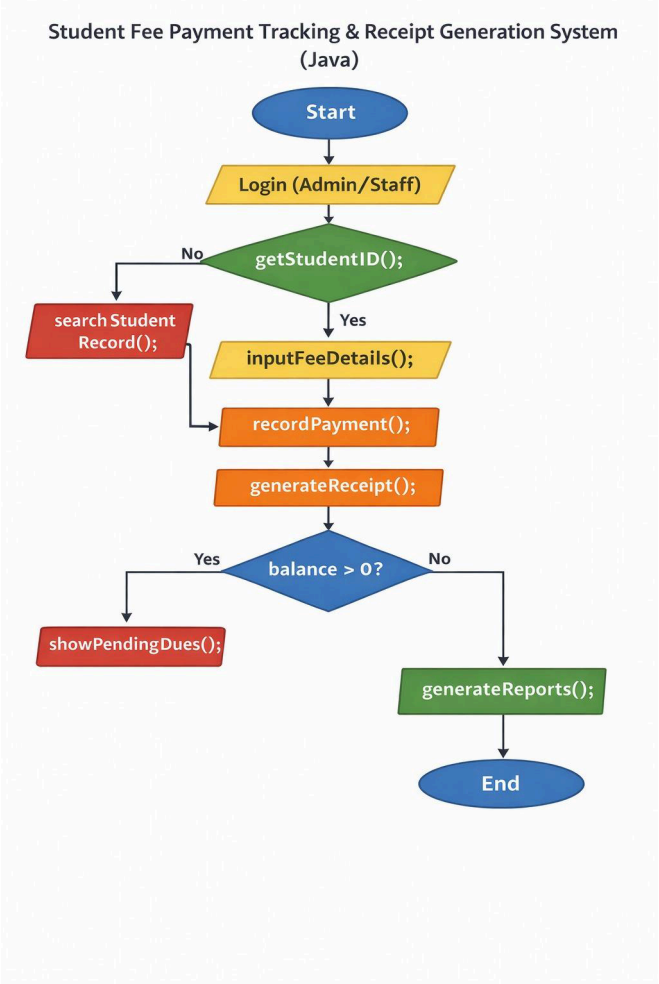
One of the major problems is data inaccuracy, as manual entry increases the chances of human errors such as incorrect calculations, missing entries, or duplication of records. Another issue is time consumption, since searching for student records or updating fee details takes a lot of time and effortt

4.6. Algorithm

Step-by-step procedure:

- Start the program
- Display menu
- Enter student details
- Enter total fee
- Enter paid amount
- Calculate balance
- Generate receipt
- Display result
- Ask for next operation (optional)
- End program

4.7. Flowchart



5. SOURCE CODE

```
import java.util.Scanner;
class StudentFeeSystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int choice;
        int id;
        String name;
        double totalFee, paid, balance;

        do {
            System.out.println("\n--- Student Fee System ---");
            System.out.println("1. Enter Student Details & Pay Fee");
            System.out.println("2. View Receipt");
            System.out.println("3. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();

            switch (choice) {

                case 1:
```

```
System.out.print("Enter Student ID: ");
id = sc.nextInt();

System.out.print("Enter Student Name: ");
name = sc.next();

System.out.print("Enter Total Fee: ");
totalFee = sc.nextDouble();

System.out.print("Enter Paid Amount: ");
paid = sc.nextDouble();

balance = totalFee - paid;

System.out.println("\nPayment Recorded Successfully!");
System.out.println("\n--- Receipt ---");
System.out.println("Student ID: " + id);
System.out.println("Name: " + name);
System.out.println("Total Fee: " + totalFee);
System.out.println("Paid Amount: " + paid);
System.out.println("Balance: " + balance);

if (balance > 0) {
    System.out.println("Status: Pending Fees");
} else {
    System.out.println("Status: Fully Paid");
```

```
                }
                break;

            case 2:
                System.out.println("\nReceipt already shown after payment.");
                break;

            case 3:
                System.out.println("Exiting System...");
                break;

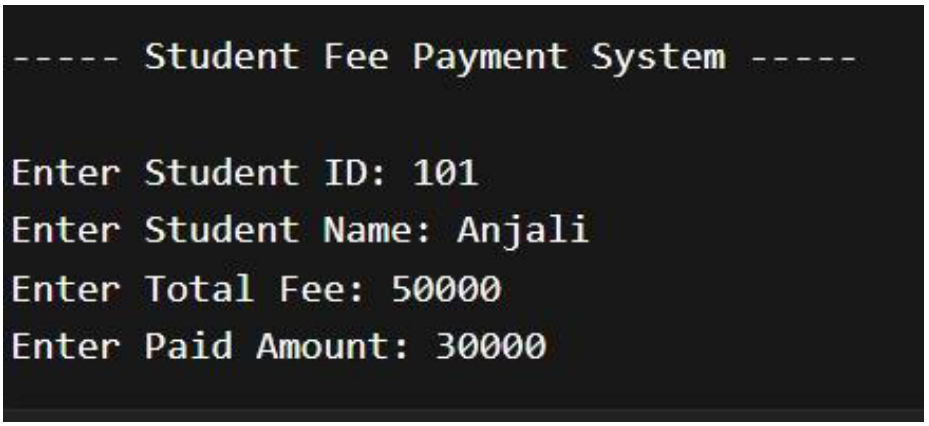
            default:
                System.out.println("Invalid choice!");
        }

    } while (choice != 3);

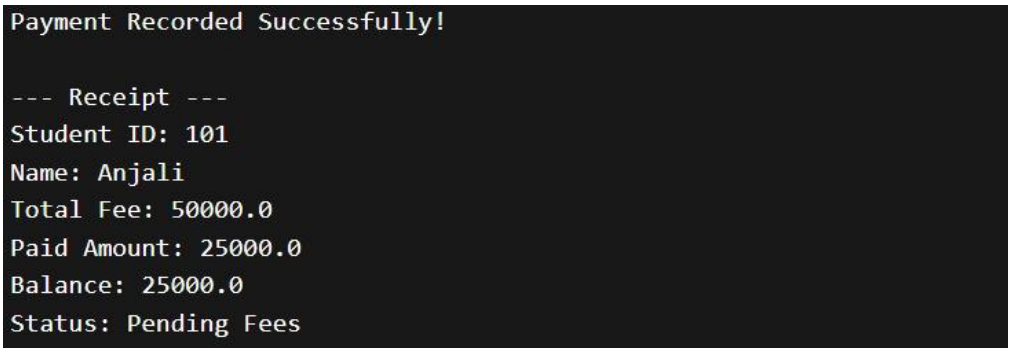
    sc.close();
  }
}
```

6. INPUT AND OUTPUT SCREENS

Input screen:



Output screen:



7. DISCUSSION & CONCLUSION

7.1 Discussion

The Student Fee Payment Tracking and Receipt Generation System was developed to simplify and automate the process of managing student fee payments. The system replaces the traditional manual method with a computerized approach, making the process faster and more efficient.

During the development of this project, various Core Java concepts such as classes, objects, loops, conditional statements, and user input handling were successfully implemented. The system allows users to enter student details, record fee payments, and automatically calculate the remaining balance. It also generates a clear and structured receipt, which improves the overall professionalism of the fee management process.

7.2 Conclusion

The Student Fee Payment Tracking and Receipt Generation System has been successfully developed using Java to simplify the process of managing student fee payments. The system effectively replaces manual methods by providing an automated solution for recording payments, calculating balances, and generating receipts.The **Student Fee Payment Tracking and Receipt Generation System** proves to be a highly practical and impactful solution for modern educational institutions. By automating fee collection and record management, the system significantly reduces the dependency on manual paperwork, which is often time-consuming and prone to errors.

One of the major advantages of this system is its ability to maintain accurate and organized records of all transactions. This helps administrators quickly retrieve student fee details, check pending payments, and generate reports for decision-making. The instant receipt generation feature ensures professionalism and builds trust with students and parents.

From a technical perspective, implementing the system using Java enhances reliability, security, and scalability. The modular design allows future expansion, such as adding features like online payment gateways, SMS/email notifications, and integration with student management systems

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

- https://www.oracle.com/java
- https://www.geeksforgeeks.org
- https://www.w3schools.com
- https://www.javatpoint.com